LOYOLA
UNIVERSITY MARYLAND

Economics 420: Econometrics
Spring Semester 2025
Maryland Hall 440
Tuesdays and Thursdays 9:25 AM - 10:40 AM

Instructor: Dr. Kerry Tan
Email: kmtan@loyola.edu
Zoom: https://loyola.zoom.us/my/kmtan
Office: Sellinger Hall 321
Office Hour: By appointment

Course Website: http://moodle.loyola.edu/
Lecture slides, homework, data sets, and other course materials are posted on Moodle.

Course Overview:
This course exposes you to the statistical techniques that economists use for estimating and testing economic relationships. It builds on topics covered in Business Statistics (EC 220) and analyzes these topics in greater breadth and depth. There will be significant emphasis on the use of Stata, a data analysis and statistical program commonly used in academia and the private sector. You will learn how to use Stata to organize data, run regressions, and interpret the results. By the end of this course, you should confidently include your experience with Stata as a skill on your resume and LinkedIn profile, which will help you be more competitive on the job market.

Course Prerequisites:
This course comes with three prerequisites: 1) Microeconomic Principles (EC 102), 2) Macroeconomic Principles (EC 103), and 3) either Business Statistics (EC 220) or Introduction to Statistics (ST 210). I will assume that you are competent in topics covered in these courses and will not reserve significant class time to review any material that was taught to you in these classes.

Textbook:
Jeffrey Wooldridge
Introductory Econometrics: A Modern Approach (6th Edition), Cengage (2016)
ISBN: 9781305270107

Stata:
Students can access Stata through the Apporto Virtual Lab link on Moodle, which will open a tab in your browser that resembles a full Windows desktop:

- There is a 15-minute inactivity timer, which registers keyboard and mouse activity. After 15 minutes of inactivity, your session will be disconnected. After being disconnected, you have another 15 minutes to reconnect to the session before you are signed out. Once signed out, all applications will close and any unsaved work will be lost.
- If you close the browser window or tab, you will be signed out of the virtual lab and all applications will close and any unsaved work will be lost.
- The Upload and Download utility in the toolbar lets you upload and download files onto the desktop or documents folder in the Apporto environment. Click here for instructions on using the upload and download utility

Grading Policy:
Grades for this course will be assigned as follows:
Exam 1: 25% (February 18)
Exam 2: 25% (March 25)
Exam 3: 25% (April 24)
Homework Assignments: 20%
Surveys: 5%

Exams:
There will be 3 in-class exams, consisting of 15 multiple choice questions and 4 word problems. Exam 1 will only cover Chapters 1, 2, and 3. Exam 2 will only cover Chapters 4, 6, and 7. Exam 3 will only cover Chapters 13, 14, and 15.

Makeup exams will *not* be given without documented medical emergency or family emergency. Travel plans are *not* an emergency. It is your responsibility to plan around scheduled exam times.

Homework Assignments:
3 homework assignments will be assigned in this course. The homework (and solutions) will be posted on the course website. Each homework assignment will be worth 25 points and will be graded based on accuracy. Students are encouraged to collaborate in small groups, but each student must turn in his or her own assignment. Students must hand in their homework in class on the assigned due date. No late homework will be accepted without a valid excuse.

Surveys:
At the end of each week, you will fill out a survey on Moodle that provides feedback on your understanding of the course material. I will address your survey responses at the beginning of each class on the following Tuesday. Since surveys are intended to be low-effort/high-reward assignments, I do not give extra credit, especially at the end of the semester.

Grading Scale:
Generally, final letter grades will follow the typical structure of: A (93-100), A- (90-92.99), B+ (87-89.99), B (83-86.99), B- (80-82.99), C+ (77-79.99), C (73-76.99), C- (70-72.99), D+ (67-69.99), D (60-66.99), and F (below 60). However, this breakdown is not set in stone and can vary according to the distribution of raw grades.

Students with Disabilities:
If you are a student with a disability requesting accommodations for this course, please submit your Faculty Notification Email (FNE) request to Disability Support Services as soon as possible. After the FNE has been sent, please schedule a brief meeting with me during my office hours to discuss accommodations. If you need additional academic accommodations because of remote learning, please contact your DSS advisor immediately.

If you are not registered with DSS, but feel you might need accommodations or modifications for my course because of a physical, medical, emotional, or learning impairment, you must register with Disability Support Services. To register with DSS, or if you have questions about disability accommodations, please contact the Disability Support Services Office at dss@loyola.edu.

Academic Misconduct:
All students are expected to understand the meaning of the Loyola Honor Code, which states:

"All students of the Loyola Community have been equally entrusted by their peers to conduct themselves honestly on all academic assignments.

The students of this University understand that having collective and individual responsibility for the ethical welfare of their peers exemplifies a commitment to the community. Students who submit materials that are the products of their own minds demonstrate respect for themselves and the community in which they study."

Ignorance of the Code is not a valid reason for committing an act of academic dishonesty. The following constitute violations of the Code: cheating, stealing, lying, forgery, plagiarism, and the failure to report a violation. Violations of the Honor Code will be handled by the Honor Council.

Course Schedule:

Date	**Chapter Covered**
Jan. 14, 16	EC 220 Refresher
Jan. 21	No Class: Loyola CFO Interviews
Jan. 23	Ch 1 - The Nature of Econometrics and Economic Data
Jan. 28, 30	Ch 2 - The Simple Regression Model
Feb. 4, 11	Ch 3 - Multiple Regression Analysis: Estimation
Feb. 6	No Class: Prof. Tan's Wedding
Feb. 13	Review for Exam 1
Feb. 18	EXAM 1
Feb. 20, 25	Ch 4 - Multiple Regression Analysis: Inference
Feb. 27, Mar. 11	Ch 6 - Multiple Regression Analysis: Further Issues
Mar. 4, 6	No Classes: Spring Break
Mar. 13, 18	Ch. 7 - Multiple Regression Analysis with Quantitative Information: Binary (or Dummy) Variables
Mar. 20	Review for Exam 2
Mar. 25	EXAM 2
Mar. 27, Apr. 1	Ch 13 - Pooling Cross Sections across Time: Simple Panel Data Methods
Apr. 3, 8	Ch 14 - Advanced Panel Data Methods
Apr. 10, 15	Ch 15 - Instrumental Variables Estimation and Two Stage Least Squares
Apr. 17	No Class: Easter Break
Apr. 22	Review for Exam 3
Apr. 24	EXAM 3